

Lesson 6 Lesson-End Project

Designing Solution for High Availability

Project Agenda: To design a solution for high availability for the given scenario

Description: You have been given a project to choose the right candidate service for a relational database that can be used for an OLTP (Online Transaction Processing) application. You need to ensure that during an adverse event like a catastrophic failure or natural disaster, the users are still able to read the data. You need to configure geo-replication so that you have a secondary database in place.

Tools required: Azure account with administrator access

Prerequisites: None

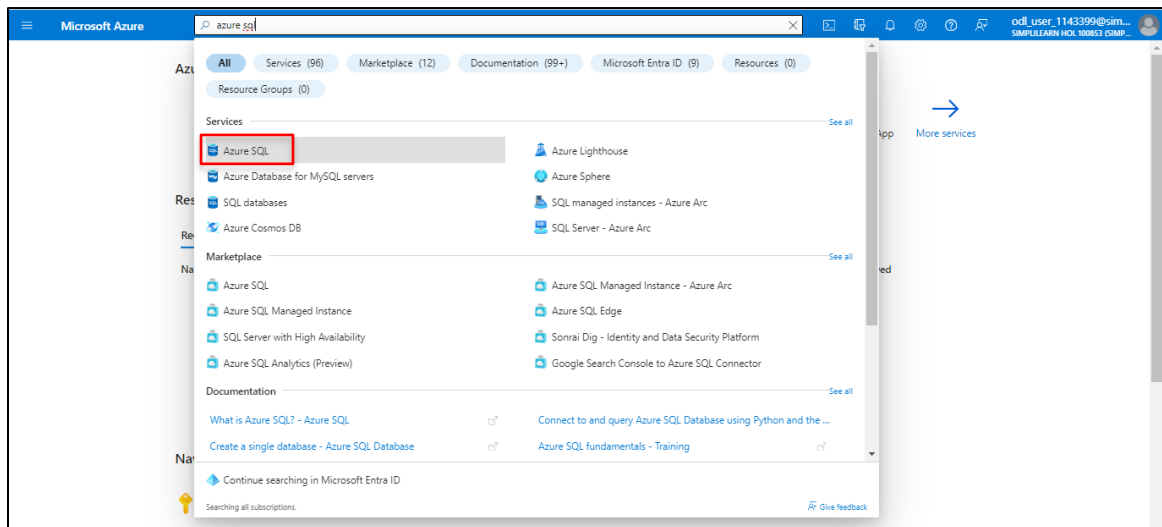
Expected deliverables: Add a screenshot of geo-replication for the SQL database created

Steps to be followed:

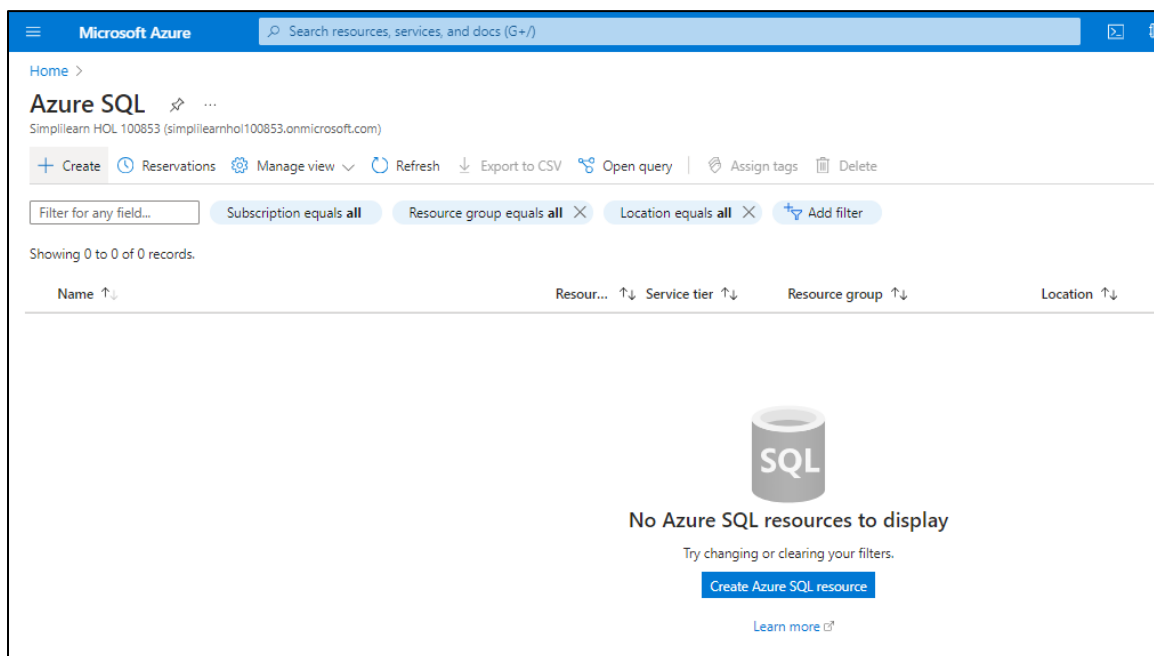
1. Configure SQL database
2. Configure geo-replication for a secondary database

Step 1: Configure SQL database

1.1 Log in to the Azure portal, search for and select **Azure SQL**



1.2 In the **Azure SQL** page, click on **+ Create**



1.3 In the **Select SQL deployment option** page, select **SQL databases** with **Single database** under **Resource type**, and click on **Create**

Microsoft Azure

Search resources, services, and docs (G+)

Home > Azure SQL >

Select SQL deployment option

Microsoft

Feedback

How do you plan to use the service?

SQL databases

Best for modern cloud applications. Hyperscale and serverless options are available.

Resource type

Single database

Create Show details

SQL managed instances

Best for most migrations to the cloud. Lift-and-shift ready.

Resource type

Single instance

Create Show details

SQL virtual machines

Best for migrations and applications requiring OS-level access. Lift-and-shift ready.

Image

Create Show details High availability

1.4 In the **Basics** tab of the **Create SQL Database** page, under **Resource group**, click on **Create new** to create a new resource group

Microsoft Azure

Search resources, services, and docs (G+)

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Basics Networking Security Additional settings Tags Review + create

Create a SQL database with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. [Learn more](#)

Want to try Azure SQL Database for free? Create a free serverless database with the first 100,000 vCore seconds, 32GB of data, and 32GB of backup storage free per month for the lifetime of the subscription. [Learn more](#)

Apply offer (Preview)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Simplilearn HOL 100853

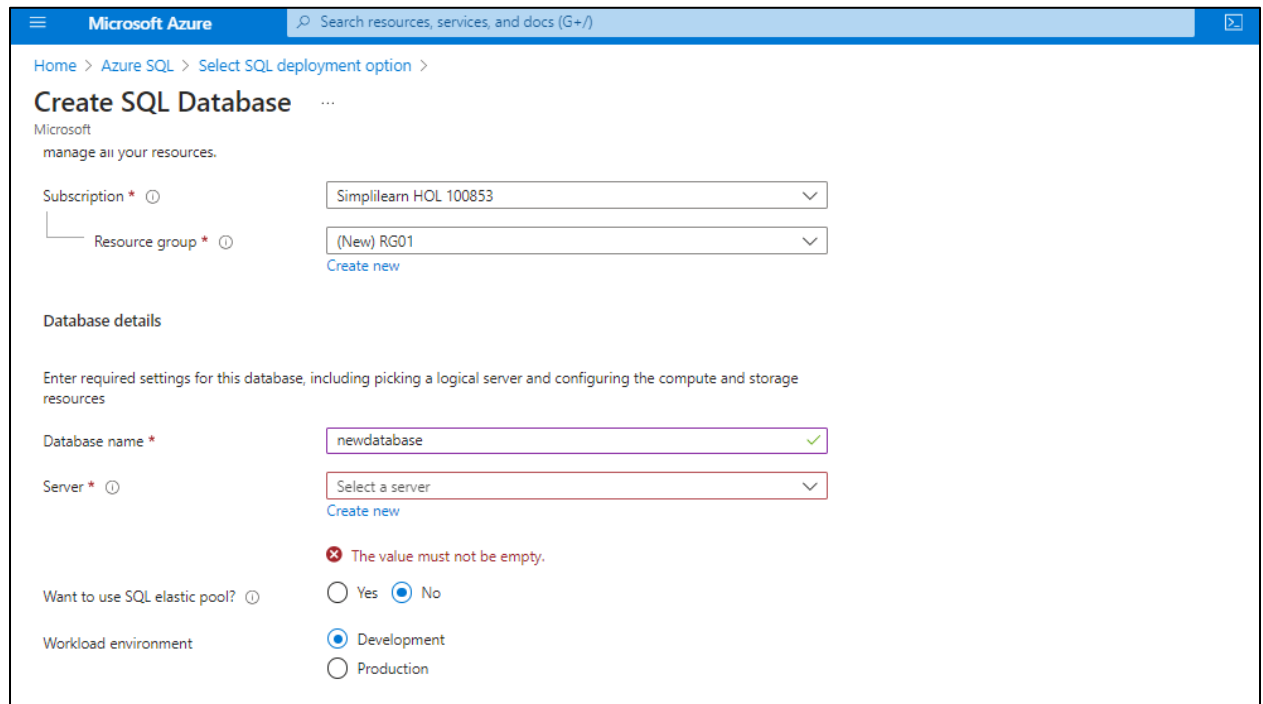
Resource group * (New) RG01

Create new

Database details

Review + create Next: Networking >

1.5 Under **Database details**, enter the **Database name**, and then click on **Create new** to select a **Server**



Microsoft Azure

Search resources, services, and docs (G+)

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft
manage all your resources.

Subscription * ⓘ Simplilearn HOL 100853

Resource group * ⓘ (New) RG01
[Create new](#)

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name * newdatabase ✓

Server * ⓘ Select a server
[Create new](#)

✖ The value must not be empty.

Want to use SQL elastic pool? ⓘ ☐ Yes ☒ No

Workload environment ☒ Development ☐ Production

1.6 In the **Create SQL Database Server** page, enter **Server name** and **Location** as shown in the screenshot below:

1.7 Next, under the **Authentication method** select **Use SQL authentication** and then enter **Server admin login**, **Password**, and **Confirm password**. Finally, click on **OK**.

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Azure SQL > Select SQL deployment option > Create SQL Database >

Create SQL Database Server

Microsoft

Authentication

i Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#), or select both SQL and Microsoft Entra authentication.

Authentication method

- ☐ Use Microsoft Entra-only authentication
- ☐ Use both SQL and Microsoft Entra authentication
- ☒ Use SQL authentication

Server admin login * azureuser ✓

Password * ✓

Confirm password * ✓

OK

1.8 Under **Compute + storage**, select **Configure database**

Microsoft Azure

Search resources, services, and docs (G+)

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Default settings provided for Development workloads. Configurations can be modified as needed.

Compute + storage * ⓘ

General Purpose - Serverless
Standard-series (Gen5), 1 vCore, 32 GB storage, zone redundant disabled

Configure database

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

Backup storage redundancy ⓘ

☐ Locally-redundant backup storage

☐ Zone-redundant backup storage

☒ Geo-redundant backup storage

Selected value for backup storage redundancy is Geo-redundant backup storage. Database backups will be geo-replicated which might impact your data residency requirements. [Learn more](#)

Review + create Next : Networking >

1.9 In the **Configure** page, enter the configurations per the requirement and click on **Apply**

Microsoft Azure

Search resources, services, and docs (G+/I)

Home > Azure SQL > Select SQL deployment option >

Configure

Feedback

Hardware Configuration

Standard-series (Gen5)

up to 80 vCores, up to 240 GB memory

Change configuration

Max vCores

2

Min vCores

0.5 vCores

2.05 GB MIN MEMORY

6 GB MAX MEMORY

Auto-pause delay

The database automatically pauses if it is inactive for the time period specified here, and automatically resumes when database activity recurs. Alternatively, auto-pausing can be disabled.

Enable auto-pause

Days

Hours

Minutes

0

1

0

Data max size (GB)

20

Apply

NOTES

1 Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

PLEASE CONTACT YOUR RESELLER

1.10 Click on **Next: Networking >**

1.11 In the **Networking** tab, choose the **Connectivity method** as the **Public endpoint**, under **Firewall rules**, set **Add current client IP address** to **Yes**. Then, click on **Next: Security >**.

Microsoft Azure

Search resources, services, and docs (G+)

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Basics

Networking

Security

Additional settings

Tags

Review + create

Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'serverdemo67' and all databases it manages. [Learn more](#)

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

Connectivity method *

No access

Public endpoint

Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

No Yes

Add current client IP address *

No Yes

Review + create

< Previous

Next : Security >

SQL

Cost summary

General Purpose (GP_Gen5_2)

Cost per GB (in --)

Max storage selected (in GB)

x 26

ESTIMATED STORAGE COST / MONTH

COMPUTE COST / VCORE SECOND

NOTES

1 Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

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1.12 In the **Security** tab, under **Enable Microsoft Defender for SQL**, select **Start free trial**. Then, click on **Next: Additional settings >**.

The screenshot shows the 'Create SQL Database' wizard in the Microsoft Azure portal, specifically the 'Security' tab. The breadcrumb trail is 'Home > Azure SQL > Select SQL deployment option >'. The page title is 'Create SQL Database' with a Microsoft logo. The navigation tabs are 'Basics', 'Networking', 'Security' (selected), 'Additional settings', 'Tags', and 'Review + create'.

Microsoft Defender for SQL

Protect your data using Microsoft Defender for SQL, a unified security package including vulnerability assessment and advanced threat protection for your server. [Learn more](#)

Get started with a 30 day free trial period, and then --/server/month.

Enable Microsoft Defender for SQL * ☒ Start free trial ☐ Not now

Ledger

Ledger cryptographically verifies the integrity of your data and detects any tampering that might have occurred. [Learn more](#)

Ledger Not configured [Configure ledger](#)

Server identity

Use system assigned and user assigned managed identities to enable central access management between this database and other Azure resources. [Learn more](#)

Cost summary

General Purpose (GP_S_Gen5_2)	
Cost per GB (in --)	--
Max storage selected (in GB)	x 26
ESTIMATED STORAGE COST / MONTH	-- ..
COMPUTE COST / VCORE SECOND ¹	-- ..

NOTES

¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

PLEASE CONTACT YOUR RESELLER

At the bottom, there are three buttons: 'Review + create' (blue), '< Previous' (disabled), and 'Next : Additional settings >' (disabled).

1.13 In the **Additional settings** tab, under **Use existing data**, select **Sample**. Then, click on **Review + create**.

Microsoft Azure

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Basics Networking Security **Additional settings** Tags Review + create

Customize additional configuration parameters including collation & sample data.

Data source

Start with a blank database, restore from a backup or select sample data to populate your new database.

Use existing data * None Backup **Sample**

AdventureWorksLT will be created as the sample database.

Database collation

Database collation defines the rules that sort and compare data, and cannot be changed after database creation. The default database collation is SQL_Latin1_General_CP1_CI_AS. [Learn more](#)

Collation SQL_Latin1_General_CP1_CI_AS

Cost summary

General Purpose (GP_S_Gen5_2)	
Cost per GB (in --)	--
Max storage selected (in GB)	x 26
ESTIMATED STORAGE COST / MONTH	--
COMPUTE COST / VCORE SECOND ¹	--

NOTES

¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

PLEASE CONTACT YOUR RESELLER

Review + create < Previous Next: Tags >

1.14 Click on **Create** after reviewing the settings

Microsoft Azure

Home > Azure SQL > Select SQL deployment option >

Create SQL Database

Microsoft

Basics Networking Security Additional settings Tags **Review + create**

Product details

SQL database by Microsoft

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Estimated cost

Storage cost -- / month + Compute cost -- / vCore second

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription	Simplilearn HOL 100853
Resource group	RG01
Region	East US
Database name	newdatabase
Server	(new) serverdemo67

Cost summary

General Purpose (GP_S_Gen5_2)	
Cost per GB (in --)	--
Max storage selected (in GB)	x 26
ESTIMATED STORAGE COST / MONTH	--
COMPUTE COST / VCORE SECOND ¹	--

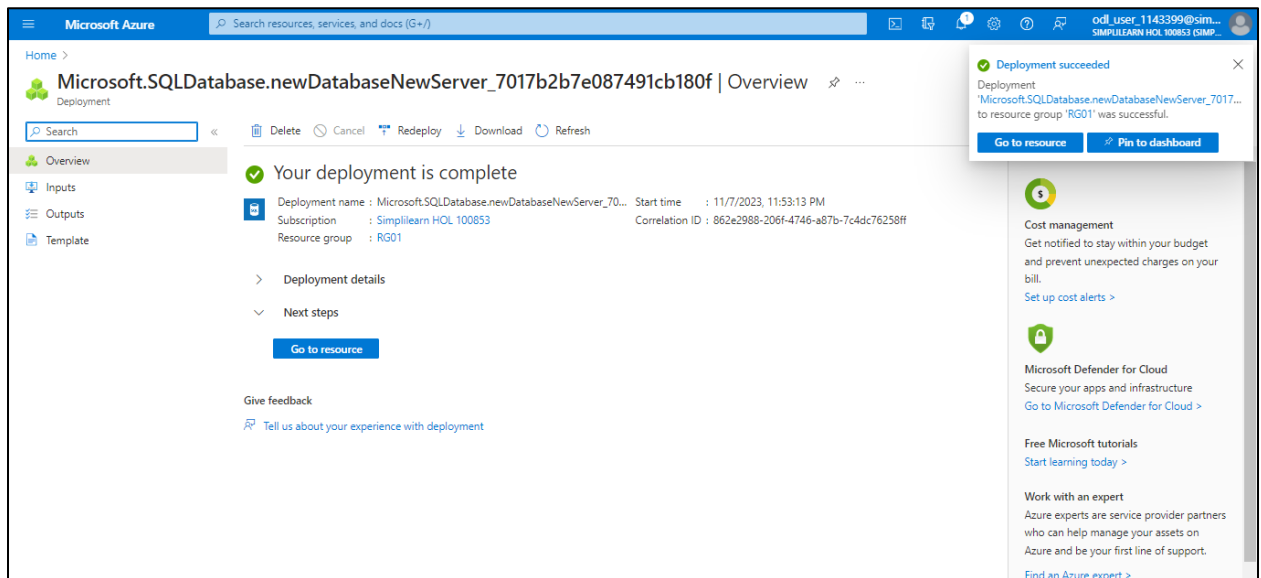
NOTES

¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

PLEASE CONTACT YOUR RESELLER

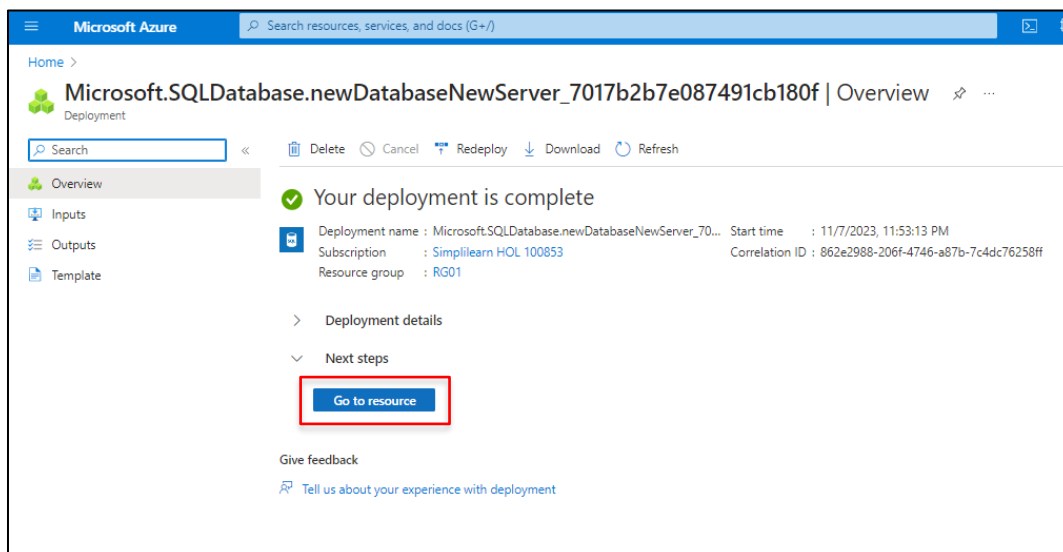
Create < Previous [Download a template for automation](#)

The Azure SQL DB is deployed as shown in the screenshot below:



Step 2: Configure geo-replication for a secondary database

2.1 Click on Go to resource



2.2 Select **Replicas** under **Data management**

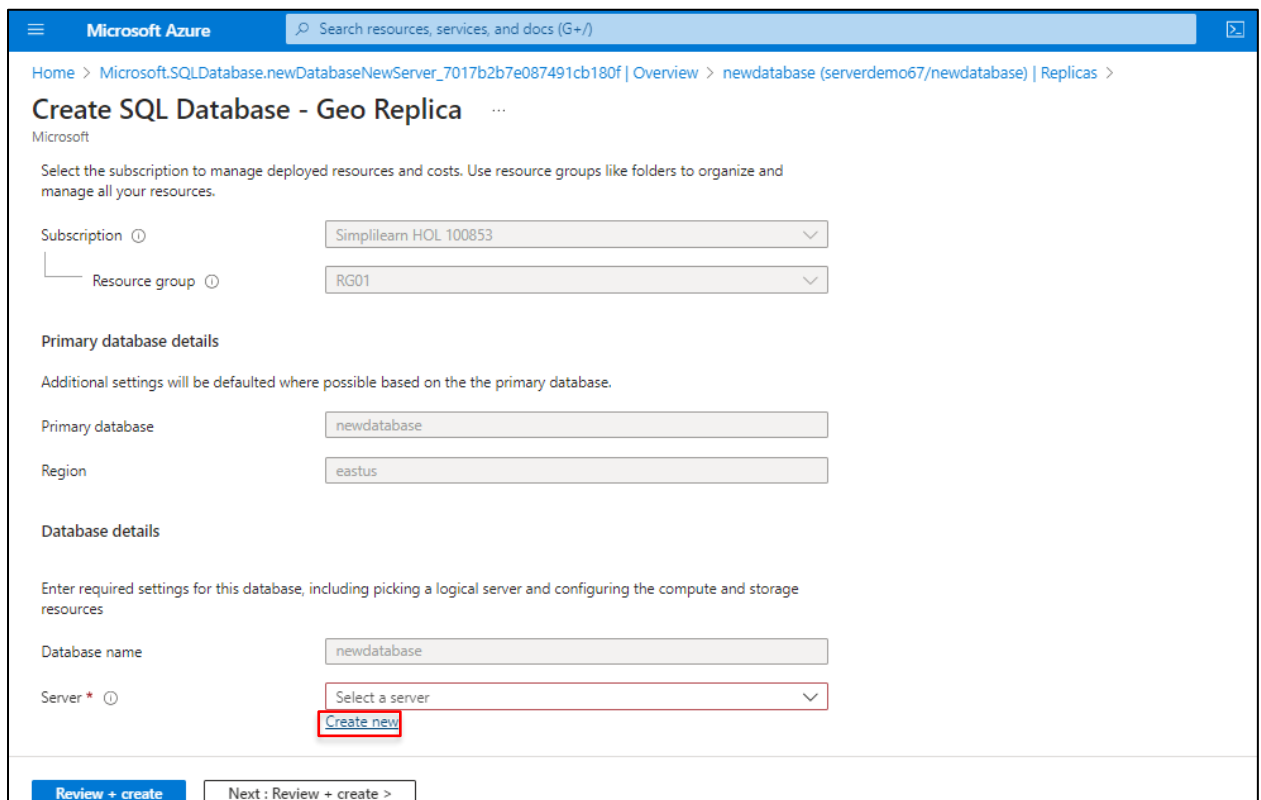
The screenshot shows the Azure portal interface for a new SQL database. The left-hand navigation pane is expanded, showing the 'Replicas' option under the 'Data management' section. The main content area displays the 'Essentials' section with details about the resource group, status, location, subscription, and tags. Below this, there are tabs for 'Getting started', 'Monitoring', 'Properties', 'Features', 'Notifications (1)', 'Integrations', and 'Tutorials'. The 'Getting started' tab is active, showing a 'Start working with your database' section with three cards: 'Configure access', 'Connect to application', and 'Start developing'.

2.3 In the **Replicas** page, click on **+ Create replica**

The screenshot shows the Azure portal interface for the 'Replicas' page of the 'newdatabase (serverdemo67/newdatabase)'. The left-hand navigation pane is expanded, showing the 'Replicas' option under the 'Data management' section. The main content area displays the 'Replicas' section with a table of replicas. The '+ Create replica' button is highlighted with a red box. Below the table, there is a message: 'Geo replicas for your database are listed below. Geo replicas reside on a different logical server from the primary and protect against regional failure'.

Name ↑↓	Server ↑↓	Region ↑↓	Failover policy ↑↓	Pricing tier
No replicas found				

2.4 In the **Create SQL Database – Geo Replica** page, under **Server**, click on **Create new**



Microsoft Azure

Search resources, services, and docs (G+/I)

Home > Microsoft.SqlDatabase.newDatabaseNewServer_7017b2b7e087491cb180f | Overview > newdatabase (serverdemo67/newdatabase) | Replicas >

Create SQL Database - Geo Replica

Microsoft

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription ⓘ Simplilearn HOL 100853

Resource group ⓘ RG01

Primary database details

Additional settings will be defaulted where possible based on the the primary database.

Primary database newdatabase

Region eastus

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

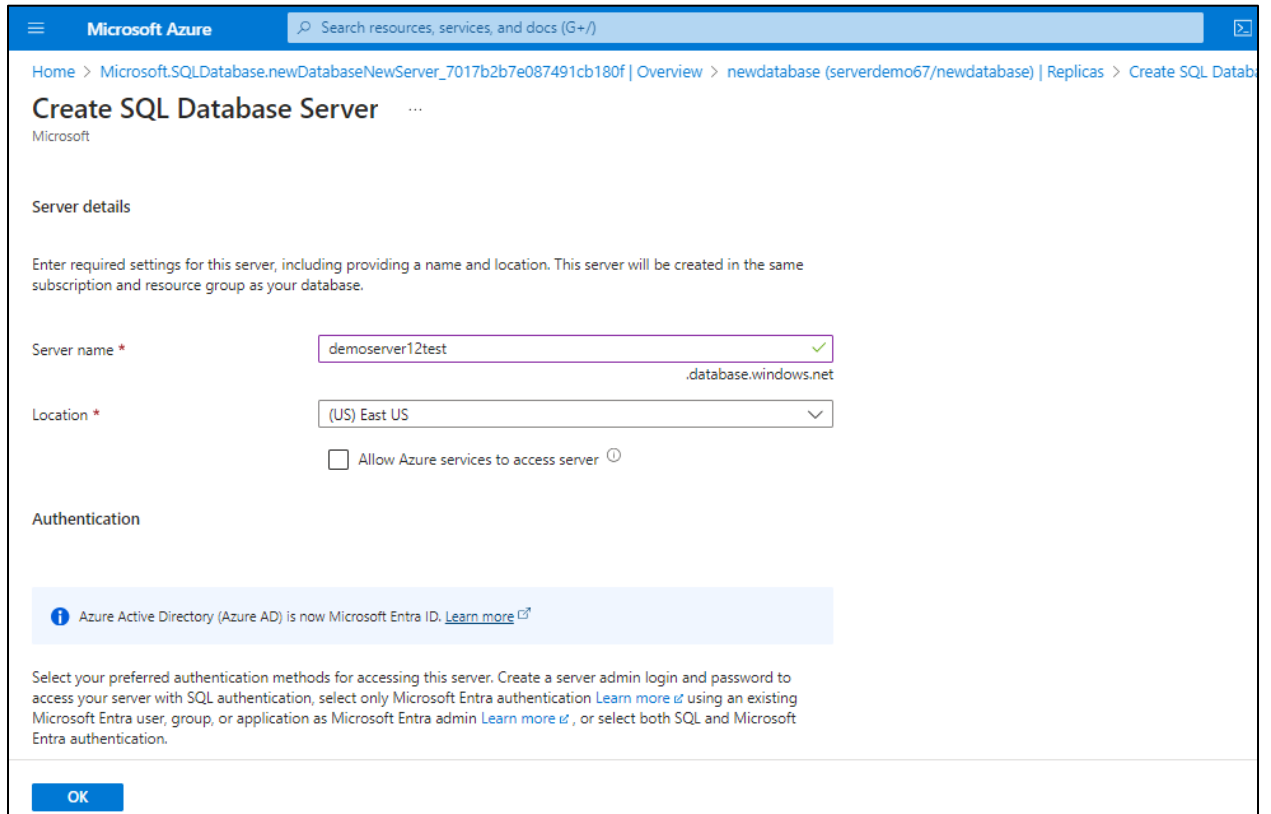
Database name newdatabase

Server * ⓘ Select a server

Create new

Review + create Next : Review + create >

2.5 In **Create SQL Database Server** page, enter the **Server name** and **Location** as shown in the screenshot below:



Microsoft Azure

Search resources, services, and docs (G+)

Home > Microsoft.SqlDatabase.newDatabaseNewServer_7017b2b7e087491cb180f | Overview > newdatabase (serverdemo67/newdatabase) | Replicas > Create SQL Database

Create SQL Database Server

Microsoft

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name * ✓
.database.windows.net

Location * ▼

☐ Allow Azure services to access server ⓘ

Authentication

i Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#) ⓘ

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) ⓘ using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#) ⓘ, or select both SQL and Microsoft Entra authentication.

OK

2.6 Now, under **Authentication method**, select **Use SQL authentication** and enter **Server admin login**, **Password**, and **Confirm password**. Then, click on **OK**.

Microsoft Azure Search resources, services, and docs (G+)

Home > Microsoft.SQLDatabase.newDatabaseNewServer_7017b2b7e087491cb180f | Overview > newdatabase (serverdemo67/newdatabase) | Replicas > Create SQL

Create SQL Database Server

Microsoft

Authentication

i Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#), or select both SQL and Microsoft Entra authentication.

Authentication method

- ☐ Use Microsoft Entra-only authentication
- ☐ Use both SQL and Microsoft Entra authentication
- ☒ Use SQL authentication

Server admin login *

azureuser ✓

Password *

..... ✓

Confirm password *

..... ✓

OK

2.7 Click on **Review + create**

The screenshot displays the Microsoft Azure portal interface for creating a new SQL Database with Geo Replicas. The breadcrumb trail indicates the path: Home > Microsoft.SqlDatabase.newDatabaseNewServer_7017b2b7e087491cb180f | Overview > newdatabase (serverdemo67/newdatabase) | Replicas >. The main heading is 'Create SQL Database - Geo Replica' with a Microsoft logo. Below this, a note states: 'Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.'

The configuration form includes the following sections and fields:

- Subscription:** A dropdown menu showing 'Simplilearn HOL 100853'.
- Resource group:** A dropdown menu showing 'RG01'.
- Primary database details:**
 - A note: 'Additional settings will be defaulted where possible based on the the primary database.'
 - Primary database:** A text input field containing 'newdatabase'.
 - Region:** A dropdown menu showing 'eastus'.
- Database details:**
 - A note: 'Enter required settings for this database, including picking a logical server and configuring the compute and storage resources'.
 - Database name:** A text input field containing 'newdatabase'.
 - Server *:** A dropdown menu showing '(new) demoserver12test (East US)' with a 'Create new' link below it.

At the bottom of the form, there are two buttons: 'Review + create' (highlighted with a red box) and 'Next : Networking >'.

2.8 Click **Create** to add the secondary database

Microsoft Azure

Search resources, services, and docs (G+)

[Home](#) > [Microsoft.SQLDatabase.newDatabaseNewServer_7017b2b7e087491cb180f](#) | [Overview](#) > [newdatabase \(serverdemo67/newdatabase\)](#) | [Replicas](#) >

Create SQL Database - Geo Replica

Microsoft

SQL database

by Microsoft

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Estimated cost

Storage cost -- -- / month + Compute cost -- -- / vCore second

Terms

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Basics

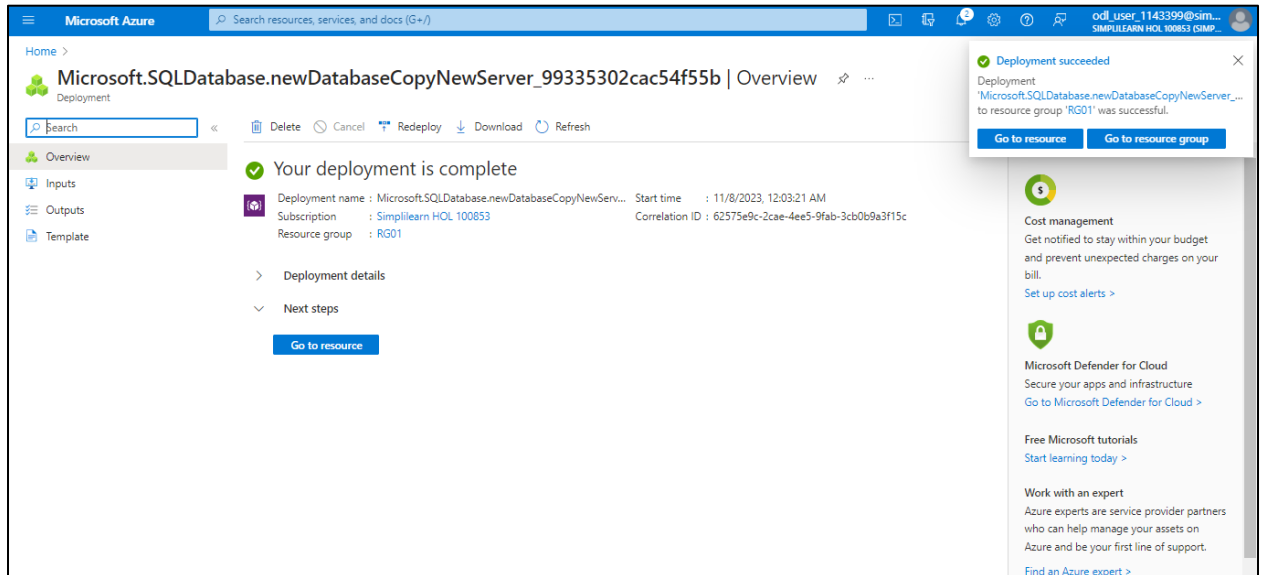
Subscription	Simplilearn HOL 100853
Resource group	RG01
Primary database	newdatabase
Region	East US
Database name	newdatabase
Server	(new) demoserver12test
Authentication method	SQL authentication
Server admin login	azureuser
Compute + storage	General Purpose - Serverless: Standard-series (Gen5), 2 vCores, 32 GB storage, zone redundant disabled
Database level customer-managed key	Not configured
Database level user assigned managed	Not configured

Create

< Previous

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A secondary database will be created in a few minutes as shown in the screenshot below:



By following these steps, you have successfully designed a high availability solution for the scenario stated in the description.